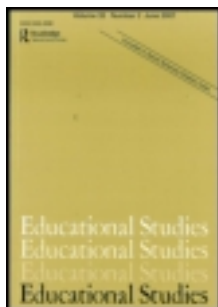


This article was downloaded by: [Alexander Häfner]

On: 25 March 2014, At: 06:01

Publisher: Routledge

Informa Ltd Registered in England and Wales Registered Number: 1072954 Registered office: Mortimer House, 37-41 Mortimer Street, London W1T 3JH, UK



Educational Studies

Publication details, including instructions for authors and subscription information:

<http://www.tandfonline.com/loi/ceds20>

Avoiding procrastination through time management: an experimental intervention study

Alexander Häfner^a, Verena Oberst^a & Armin Stock^a

^a Adolf-Würth-Center for the History of Psychology, University of Würzburg, Würzburg, Germany

Published online: 21 Mar 2014.

To cite this article: Alexander Häfner, Verena Oberst & Armin Stock (2014): Avoiding procrastination through time management: an experimental intervention study, Educational Studies, DOI: [10.1080/03055698.2014.899487](https://doi.org/10.1080/03055698.2014.899487)

To link to this article: <http://dx.doi.org/10.1080/03055698.2014.899487>

PLEASE SCROLL DOWN FOR ARTICLE

Taylor & Francis makes every effort to ensure the accuracy of all the information (the "Content") contained in the publications on our platform. However, Taylor & Francis, our agents, and our licensors make no representations or warranties whatsoever as to the accuracy, completeness, or suitability for any purpose of the Content. Any opinions and views expressed in this publication are the opinions and views of the authors, and are not the views of or endorsed by Taylor & Francis. The accuracy of the Content should not be relied upon and should be independently verified with primary sources of information. Taylor and Francis shall not be liable for any losses, actions, claims, proceedings, demands, costs, expenses, damages, and other liabilities whatsoever or howsoever caused arising directly or indirectly in connection with, in relation to or arising out of the use of the Content.

This article may be used for research, teaching, and private study purposes. Any substantial or systematic reproduction, redistribution, reselling, loan, sub-licensing, systematic supply, or distribution in any form to anyone is expressly forbidden. Terms & Conditions of access and use can be found at <http://www.tandfonline.com/page/terms-and-conditions>

Avoiding procrastination through time management: an experimental intervention study

Alexander Häfner*, Verena Oberst and Armin Stock

Adolf-Würth-Center for the History of Psychology, University of Würzburg, Würzburg, Germany

(Received 14 February 2013; final version received 25 February 2014)

The purpose of this study was to examine the effects of a short-term time management intervention on procrastination. Procrastination is a serious issue for many students and associated with different negative consequences, such as anxiety or low grades. As procrastination is described as a self-regulatory failure, a training programme focussing on self-regulatory skills might be helpful. We randomly assigned 96 students to a treatment ($n=47$) and a control group ($n=49$). As hypothesised, participants of the control group showed procrastination: they spent more time on a self-selected, important academic task right before the deadline was reached compared to the weeks before. The participants of the treatment group, instead, allocated their work time more equally and did not show procrastination. Therefore, the explored time management intervention prevents from a serious time management problem.

Keywords: time management; procrastination; planning; training programme

Steel (2007) defined “that to procrastinate is to voluntarily delay an intended course of action despite expecting to be worse off for the delay” (66). Procrastination is quite common among students: prevalence estimates concerning serious procrastination vary between about 27 and 60% for academic tasks such as writing term papers, studying for exams or reading weekly assignments (e.g. Beswick, Rothblum, and Mann 1988; Onwuegbuzie 2004; Rothblum, Solomon, and Murakami 1986; Solomon and Rothblum 1984). Procrastination has been described as a common but dysfunctional phenomenon among students (e.g. Fritzsche, Young, and Hickson 2003; Moon and Illingworth 2005). In several studies, in the last decades, procrastination was associated with depression (Solomon and Rothblum 1984; Van Eerde 2003), guilt (Pychyl, Lee, Thibodeau, and Blunt 2000), anxiety (Fritzsche, Young, and Hickson 2003; Rothblum, Solomon, and Murakami 1986), as well as low grades (Fritzsche, Young, and Hickson 2003; Klassen, Krawchuk, and Rajani 2008; Moon and Illingworth 2005; Van Eerde 2003). In their meta-analysis, Van Eerde found moderate negative correlations between procrastination and two indicators of academic performance: course grade ($r=-0.17$) and grade point average ($r=-0.28$). Furthermore, accelerated learning before a deadline has been found less successful than studying at an even pace (Ariely and Wertenbroch 2002; Perrin et al. 2011). Consequently, procrastination can be seen as a serious issue among students.

*Corresponding author. Email: Alexander.Haefner@wueth-industrie.com

Procrastination behaviour before a deadline can be best described with curvilinear functions: only little time is invested to complete a task until it becomes urgent, although there might have been weeks or months before the deadline to work on the task (Dewitte and Schouwenburg 2002; Howell et al. 2006; Koch and Kleinmann 2002; Moon and Illingworth 2005; Schouwenburg and Groenewoud 2001). This phenomenon has also been called deadline rush (Koch and Kleinmann 2002; König and Kleinmann 2005). A typical example of this kind of procrastination is the amount of time students invest in preparation for a test: the time spent follows a hyperbolic function with a strong increase of invested time right before the deadline (e.g. Howell et al. 2006; König and Kleinmann 2005). As Moon and Illingworth described it, “following a curvilinear trajectory over time, academic procrastinators tend to delay working until the last minute” (300). Nevertheless, longitudinal studies examining actual learning behaviour before a deadline are scarce (e.g. Dewitte and Schouwenburg 2002). Dewitte and Schouwenburg examined the actual study activities of students during 11 weeks before an exam. As expected the amount of time spent studying could be best described by a hyperbolic curve with most time invested in the last week before the exam.

Procrastination has been described “as a failure of self-regulation” (Howell et al. 2006, 1520). When positive outcomes of academic preparation are far away in the future, it seems to be difficult for students to transfer their good intentions into actions, to resist temptations and persist (e.g. Howell and Watson 2006; Howell et al. 2006; König and Kleinmann 2005). A lack of appropriate self-regulatory skills among students has been discussed as major reason for procrastination (e.g. Howell et al. 2006; Klassen, Krawchuk, and Rajani 2008; Wolters 2003). Howell and Watson, for example, showed negative associations between procrastination and the use of cognitive and meta-cognitive strategies. In regression analyses, cognitive and meta-cognitive strategy usage was a predictor of procrastination. Cognitive and meta-cognitive strategies cover behaviour such as rehearsal, elaboration, planning, monitoring and regulating. Furthermore, Howell and Watson found strong associations between disorganisation and procrastination underlining the assumption of procrastination as self-regulation failure.

Whereas several studies have described and examined potential causes and consequences of procrastination, research concerning interventions to avoid procrastination is scarce. Ariely and Wertenbroch (2002) showed positive effects of evenly paced deadlines on procrastination and task performance of students in a set of experimental studies. Other studies have also presented evidence for a positive impact of deadlines on procrastination (Lamwers and Jazwinski 1989; Wesp 1986). Possibly setting deadlines fosters student’s self-regulatory skills.

A popular claim to improve self-regulatory skills is to learn and practice better time management (Lynch 2008; Maguire, Evans, and Dyas 2001; Unsworth and Kauter 2008). “It is argued that universities could and should play a more active role in helping first-year students to make sense of time management” (Van der Meer, Jansen, and Torenbeek 2010, 777). Time management behaviour can be described as a combination of different strategies to achieve one’s goals over time (Claessens et al. 2007; Häfner and Stock 2010). Time assessment, planning and monitoring one’s daily work are essential elements of time management behaviour (Claessens et al. 2007; Häfner and Stock 2010).

Although fostering time management skills has been widely recommended, we found no study that examined the effects of a time management training programme

on procrastination among students. It is our aim to address this research gap. The contribution of our study is threefold: (a) we want to extend the scarce longitudinal literature on actual study behaviour before a deadline, (b) we evaluated the effectiveness of a time management training programme with respect to procrastination and (c) we tested a rather short-term intervention of about 2 h, which, if effective, could be implemented at university with little investment of time. Our main research question can be stated as follows: Is it possible to avoid procrastination through a time management intervention?

As design we used a randomised controlled trial with a time management training (treatment group) and an active control group (control group). Participants of the control group attended an alternative training programme, which should have no effect on procrastination. We assume that time management behaviour diminishes or even prevents procrastination. Through time management behaviour, goal-directed activities should be easily implemented, even if the deadline is still in the distant future. The rationale is that time management behaviour fosters the initiation of goal-directed behaviour, increases persistence, helps to resist distractions and adapts one's strategy in an adequate manner while striving for goal achievement. Therefore, time management behaviour should help to prevent procrastination through volitional processes.

Our hypotheses can be stated as follows: firstly, participants of the control group are expected to show a strong increase of time spent on a self-selected, important task in the fourth week after training, shortly before the deadline is reached. This assumption is derived from former studies concerning student procrastination before a deadline (e.g. Dewitte and Schouwenburg 2002). Secondly, participants of the treatment group should show no increase in the fourth week, due to the time management intervention. Thirdly, the overall amount of time spent on the self-selected, important task should not differ between the two groups. It just should be allocated in different ways. The procrastinating student's should try to compensate the missing time for learning in the last days before the deadline, as described in former studies (e.g. König and Kleinmann 2005; Moon and Illingworth 2005).

Method

Design

We used a randomised controlled trial, with an experimental training (treatment group) and an active control group (control group). Trainees of the active control group attended an alternative training programme of the same length and comparable structure as the experimental training group, but with different content. While participants of the experimental training group practiced time management behaviour, participants of the active control group collected and discussed time management problems. The other parts of the training programme did not differ. The use of such an active control group with the variation of a specific part allows the exploration of effects of the specific time management intervention.

Sample, procedure and measures

The sample consisted of 96 undergraduate students of a middle-sized German university. Trainees were invited to take part in the study via email newsletters and information boards. They got no incentives for taking part in the study. Trainees

were told that the training programme was designed to improve their time management skills and to avoid procrastination. They had a mean age of 23.60 years ($SD=2.78$) and studied different subjects. About half of the trainees were female (about 49%). We randomly assigned them to the experimental training group ($n=47$) and the active control group ($n=49$). For the random allocation, we used no computer programme, but put the students manually by chance in one of the two training conditions.

At the beginning of the training courses, participants selected an important task they wanted to work on for the next four weeks. We instructed trainees to select a task with a deadline after about 4 weeks. Nearly all trainees (91%) chose the preparation for an exam or writing a thesis (6%). All deadlines were external deadlines in the end of the semester, shortly after the measurement period. After taking part in the training programme, participants recorded the time spent on their self-selected, important task for each day in a diary over a time period of 4 weeks. In the diary, they had a sheet for every day with time intervals of 20 min. They were instructed to mark each time interval they used working on their task. They were asked to do this on their own at home every evening. Ninety-six participants were trained (treatment group: 47; control group: 49). Fifty-eight trainees (treatment group: 29; control group: 29) filled in the diary in the first week and second week (treatment group: 30; control group: 28; for a response rate of 60%), 52 in the third week (treatment group: 25; control group: 27; for a response rate of 54%) and 48 in the fourth week (treatment group: 25; control group: 23; for a response rate of 50%). As can be seen some participants who filled out their diaries in the first week left the study later on, while some participants who did not sent back their diaries in the first week filled out their diaries in later weeks. One participant in the treatment group was excluded after checking for outlying numbers larger than three standard deviations. For our further analysis, we could use 45 complete data sets with 22 in the treatment group and 23 in the control group. The drop-out rate did not differ between groups. Both training interventions were conceptualised as short interventions, lasting for 4 h. The specific part concerning time management behaviour took about 2 h.

In the experimental training group, participants received some information about procrastination and practiced time management behaviour. Participants in the active control group received exactly the same training intervention except for the part concerning time management behaviour. Instead of practicing time management behaviour, this group collected and discussed personal time management problems. They just talked about their problems, but neither got special guidance by the trainer concerning time management, nor practiced time management behaviour.

As aspects of time management behaviour, trainees in the experimental training group developed a strategy for reaching their goal, identified helpful proximal steps, anticipated possible obstacles and ways to overcome hindrances and imagined how to begin working towards goal achievement. They also practiced daily planning with a collection of daily tasks, prioritising, awareness of how much time they might need to finish their tasks and the implementation of concrete steps to reach their goal. They formed implementation intentions (Gollwitzer 1999) with the specification of concrete steps that should be realised in concrete situations: in situation X, I will do Y to achieve Z. The special time management training programme was developed with respect to psychological findings concerning strategy development and process simulation (e.g. Diefendorff and Lord 2003; Earley, Wojnaroski, and Prest 1987; Taylor et al. 1998) as well as the use of implementation intentions

(e.g. Gollwitzer 1999; Gollwitzer and Brandstätter 1997). The described strategies should be helpful to initiate goal-oriented behaviour, avoid distractions and persist until the goal is reached. In a recent paper, Häfner and Stock (2010) described the development of a time management training intervention with respect to psychological theory and research. For the current training programme, mainly the same sources and procedures were used.

All training courses were given by the same trainer as 4 h blocks in small groups up to 12 students. Both participants and the trainer were not informed about the hypotheses of the study. At the beginning of the training course, all participants were informed about the procedure and the treatment of their data. Participation was voluntary, and the confidentiality of responses was assured. The importance of correct and honest responses was emphasised.

Results

We compared the two groups in a repeated-measurement analysis of variance (ANOVA) with group membership as between-subject factor and summed up the amount of time spent on the self-selected, important task for each of the 4 weeks after training as dependent variable. Table 1 shows the average scores and standard deviations for the 4 weeks of measurement. As shown, large standard deviations resulted as some trainees invested several hours a day in their self-selected, important task, while others invested only a few minutes. The development of time spent on the self-selected, important task is illustrated in Figure 1, which shows a strong increase in the active control group (labelled control group) but not in the experimental training group (labelled treatment group) in the last week of the measurement period. Table 2 shows the results of the repeated-measurement ANOVA. As hypothesised, we found a significant time $\times$ training group interaction due to an increase of time spent on the self-selected, important task in the active control group. As Table 3 shows, the time $\times$ training group interaction occurs in the fourth week. While there were no differences between either group in the third week after training, $t(49) = 1.16$, $p = 0.254$ (two-tailed), the amount of time spent on the self-selected, important task increased in the active control group between the third and fourth point of measurement, $t(22) = 6.07$, $p < 0.001$ (two-tailed), $\epsilon' = 1.91$, and remained stable in the experimental training group, $t(22) = 1.04$, $p = 0.309$ (two-tailed). In the fourth week, we found a significant difference between both groups, $t(45) = 2.11$, $p = 0.040$ (two-tailed), $\epsilon' = 0.62$. As further expected, the overall amount of time spent on the self-selected, important task did not differ between the two groups, $t(43) = 0.33$, $p = 0.740$ (two-tailed).

Table 1. Means and standard deviations of time spent on important task for the four weeks of measurement (min).

Time of measurement	Treatment group ($n = 22$)		Control group ($n = 23$)	
	<i>M</i>	SD	<i>M</i>	SD
Week 1	819.09	636.67	726.96	420.76
Week 2	638.18	665.19	524.35	437.59
Week 3	860.91	685.43	784.35	509.40
Week 4	1034.55	873.60	1526.09	735.45
Total time all weeks	3352.73	2504.67	3561.74	1630.70

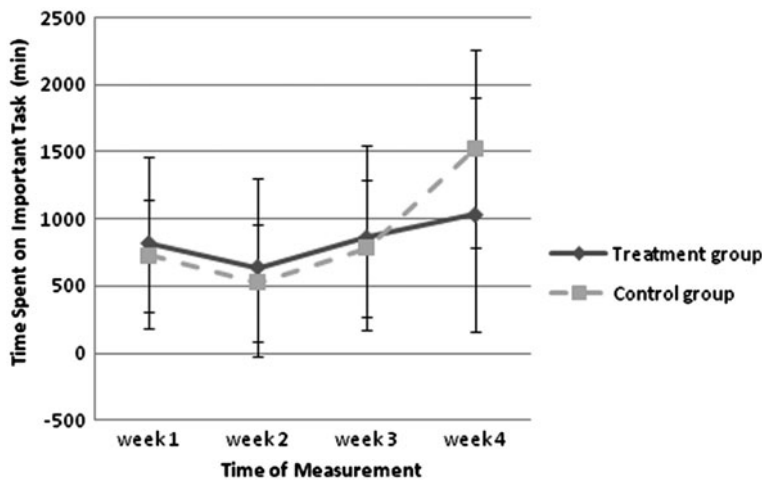


Figure 1. Amount of time spent on the self-selected, important task in the four weeks before the deadline.

Table 2. ANOVA for training group and time; t1–t4.

Source	<i>df</i>	<i>F</i>	Partial η^2	<i>p</i>
Training group	1, 43	0.11	0.00	0.74
Time	3, 41	23.11**	0.35	<0.01
Time $\times$ training group	3, 41	5.66**	0.12	<0.01

Note: All four weeks of the measurement period were included.
** $p < 0.01$.

Table 3. ANOVA for training group and time; t3 and t4.

Source	<i>df</i>	<i>F</i>	Partial η^2	<i>p</i>
Training group	1, 44	1.34	0.03	0.25
Time	1, 44	24.15**	0.35	<0.01
Time $\times$ training group	1, 44	11.52**	0.21	<0.01

Note: Third and fourth week of the measurement period were included.
** $p < 0.01$.

Discussion

As hypothesised, participants in the active control group spent more time on their self-selected, important task in the fourth week, shortly before the deadline was reached, compared to the weeks before. No increase resulted in the experimental training group. That means deadline rush occurred in the active control group but not in the experimental training group. It can be concluded that a time management training intervention, focussed on planning behaviour, prevents procrastination. Time management behaviour led to a more equal distribution of work time before the deadline. Overall, participants of the time management training group did not spend more time on their personal task, but allocated time more equally. On the one hand,

our results extend former findings of longitudinal studies on procrastination (e.g. Dewitte and Schouwenburg 2002). On the other hand, our results give new evidence concerning the effectiveness of a time management intervention.

It can be derived from several studies that planning behaviour is an effective intervention to initiate goal-directed behaviour, foster persistence and avoid distractions (e.g. Gollwitzer 1999; Gollwitzer and Brandstätter 1997). Our time management intervention was designed to strengthen such self-regulatory skills. The intervention is based on the theoretical assumption of procrastination as a self-regulatory failure (Howell et al. 2006). The positive effect of the intervention on procrastination gives further evidence for procrastination as a self-regulatory problem. While goal-directed behaviour is normally initiated right before a deadline, the taught time management behaviour changed the allocation of work time. Our results show that a rather short time management training programme is a helpful instrument for dealing with a basic self-regulatory problem.

Several, methodological aspects of our study should be mentioned: the use of randomisation and an active control group, a rather homogeneous intervention, as well as both participants and trainer being kept uninformed about the hypotheses. In former time management studies, the applied training programmes often consisted of different interventions such as self-regulatory training and relaxation exercises (e.g. Macan 1996). Consequently, the effects of such training programmes cannot be attributed to a special kind of intervention content. It remains unclear which content is the antecedent of the reported effects. Macan (1996) recommended designing and testing more homogeneous interventions in the field of time management research.

Nevertheless, some methodological and theoretical limitations still remain. We concentrated on the process of performance rather than the outcomes. We do not know if the quality of work was also affected or whether there were differences concerning the achievement of goals. On the one hand, it was a specific purpose of our study to focus on the actual use of time before a deadline. On the other hand, it might have been worthwhile to include performance measures, such as grades. Although we instructed all participants to choose a task, they wanted to spend a substantial amount of time on large standard deviations concerning the actual investment of time resulted within both groups. Such large standard deviations within one group reduce the probability of significant differences between groups. One alternative might be to define a special task for all trainees, which would allow a smaller range of invested time and reduce variances within groups. We focused on a time period of 4 weeks and found, as expected, no differences concerning the overall amount of time spent on the self-selected task. With longer-lasting projects, for example one year, there should occur differences in the overall amount of invested time between trained and untrained students, because the procrastination of working time might not be compensated through deadline rush. Our training was designed to strengthen volitional processes, such as the formation of implementation intentions. Alternative interventions should be examined with respect to procrastination. One motivational intervention to reduce procrastination might be to increase the subjective value of future outcomes through reflecting positive consequences of task achievement and negative consequences of failure. Additionally, the effectiveness of different interventions can be compared to show which strategies or which combination of strategies is the most effective one. Overall, more intervention studies should be conducted to learn more about effective ways of dealing with procrastination.

Acknowledgements

We thank the Würth Industrie Service GmbH & Co. KG for the financial support of our research. We also thank Lydia Pinneker, Julia Hartmann and Sabine Ströhle for their contributions concerning the collecting and entering of the data as well as statistical analysis and preparation of tables and figures.

Notes on contributors

Alexander Häfner is a doctor of psychology and head of human resources development at the Würth Industrie Service GmbH & Co. KG. His current research interests are effectiveness of time management training programmes, procrastination and stress prevention. His recent publications in these fields include: Time Management Training and Perceived Control of Time at Work, *Journal of Psychology* (2010); Stress Prevention Through a Time Management Training Intervention: An Experimental Study, *Educational Psychology: An International Journal of Experimental Educational Psychology* (2013).

Verena Oberst is a lecturer and researcher in the department of sports science at the University of Würzburg. She received her diploma degree in psychology from the same university in 2008. Her main research interests are effectiveness of time management training programmes and different fields in sports psychology.

Armin Stock is a doctor and professor of psychology, and is head of the Adolf-Würth-Center for the History of Psychology at the University of Würzburg. His main research interests are history of psychology, different fields in organisational psychology and cognitive psychology.

References

- Ariely, D., and K. Wertenbroch. 2002. "Procrastination, Deadlines, and Performance: Self-control by Precommitment." *Psychological Science* 13: 219–224.
- Beswick, G., E. D. Rothblum, and L. Mann. 1988. "Psychological Antecedents of Student Procrastination." *Australian Psychologist* 23: 207–217.
- Claessens, B. J. C., W. Eerde, C. G. Rutte, and R. A. Roe. 2007. "A Review of the Time Management Literature." *Personnel Review* 36: 255–276.
- Dewitte, S., and H. C. Schouwenburg. 2002. "Procrastination, Temptations, and Incentives: The Struggle between the Present and the Future in Procrastinators and Punctual." *European Journal of Personality* 16: 469–489.
- Diefendorff, J. M., and R. G. Lord. 2003. "The Volitional and Strategic Effects of Planning on Task Performance and Goal Commitment." *Human Performance* 16: 365–387.
- Earley, P. C., P. Wojnarowski, and W. Prest. 1987. "Task Planning and Energy Expended: Exploration of How Goals Influence Performance." *Journal of Applied Psychology* 72: 107–114.
- Fritzsche, B. A., B. R. Young, and K. C. Hickson. 2003. "Individual Differences in Academic Procrastination Tendency and Writing Success." *Personality and Individual Differences* 35: 1549–1557.
- Gollwitzer, P. M. 1999. "Implementation Intentions: Strong Effects of Simple Plans." *American Psychologist* 54: 493–503.
- Gollwitzer, P. M., and V. Brandstätter. 1997. "Implementation Intentions and Effective Goal Pursuit." *Journal of Personality and Social Psychology* 73: 186–199.
- Häfner, A., and A. Stock. 2010. "Time Management Training and Perceived Control of Time at Work." *Journal of Psychology* 144: 429–447.
- Howell, A. J., and D. C. Watson. 2006. "Procrastination: Associations with Achievement Goal Orientation and Learning Strategies." *Personality and Individual Differences* 43: 167–178.
- Howell, A. J., D. C. Watson, R. A. Powell, and K. Buro. 2006. "Academic Procrastination: The Pattern and Correlates of Behavioural Postponement." *Personality and Individual Differences* 40: 1519–1530.

- Klassen, R. M., L. L. Krawchuk, and S. Rajani. 2008. "Academic Procrastination of Undergraduates: Low Self-efficacy to Self-regulate Predicts Higher Levels of Procrastination." *Contemporary Educational Psychology* 33: 915–931.
- Koch, C. J., and M. Kleinmann. 2002. "A Stitch in Time Saves Nine: Behavioural Decision-making Explanations for Time Management Problems." *European Journal of Work and Organizational Psychology* 11: 199–217.
- König, C. J., and M. Kleinmann. 2005. "Deadline Rush: A Time Management Phenomenon and Its Mathematical Description." *Journal of Psychology* 139: 33–45.
- Lamwers, L. L., and C. H. Jazwinski. 1989. "A Comparison of Three Strategies to Reduce Student Procrastination in PSI." *Teaching of Psychology* 16: 8–12.
- Lynch, J. 2008. "Review of Effective PhD Candidate Series." *Higher Education Research & Development* 27: 301–303.
- Macan, T. H. 1996. "Time-management Training: Effects on Time Behaviors, Attitudes, and Job Performance." *Journal of Psychology* 130: 229–236.
- Maguire, S., S. E. Evans, and L. Dyas. 2001. "Approaches to Learning: A Study of First-year Geography Undergraduates." *Journal of Geography in Higher Education* 25: 95–107.
- Moon, S. M., and A. J. Illingworth. 2005. "Exploring the Dynamic Nature of Procrastination: A Latent Growth Curve Analysis of Academic Procrastination." *Personality and Individual Differences* 38: 297–309.
- Onwuegbuzie, A. J. 2004. "Academic Procrastination and Statistics Anxiety." *Assessment & Evaluation in Higher Education* 29: 3–19.
- Perrin, C. J., N. Miller, A. T. Haberman, J. W. Ivy, J. N. Meindl, and N. A. Neef. 2011. "Measuring and Reducing College Student's Procrastination." *Journal of Applied Behavior Analysis* 44: 463–474.
- Pychyl, T. A., J. M. Lee, R. Thibodeau, and A. Blunt. 2000. "Five Days of Emotion: An Experience Sampling Study of Undergraduate Student Procrastination." *Journal of Social Behavior and Personality* 15: 239–254.
- Rothblum, E. D., L. J. Solomon, and J. Murakami. 1986. "Affective, Cognitive, and Behavioral Differences between High and Low Procrastinators." *Journal of Counseling Psychology* 33: 387–394.
- Schouwenburg, H. C., and J. T. Groenewoud. 2001. "Study Motivation under Social Temptation; Effects of Trait Procrastination." *Personality and Individual Differences* 30: 229–240.
- Solomon, L. J., and E. D. Rothblum. 1984. "Academic Procrastination: Frequency and Cognitive-behavioral Correlates." *Journal of Counseling Psychology* 31: 503–509.
- Steel, P. 2007. "The Nature of Procrastination: A Meta-analytic and Theoretical Review of Quintessential Self-regulatory Failure." *Psychological Bulletin* 133: 65–94.
- Taylor, S. E., L. B. Pham, I. D. Rivkin, and D. A. Armor. 1998. "Harnessing the Imagination: Mental Simulation, Self-regulation, and Coping." *American Psychologist* 53: 429–439.
- Unsworth, K., and K. Kauter. 2008. "Evaluating an Earlybird Scheme: Encouraging Early Assignment Writing and Revising." *Higher Education Research & Development* 27: 69–76.
- Van der Meer, J., E. Jansen, and M. Torenbeek. 2010. "It's Almost a Mindset That Teachers Need to Change: First-year students' Need to be Inducted into Time Management." *Studies in Higher Education* 35: 777–791.
- Van Eerde, W. 2003. "A Meta-analytically Derived Nomological Network of Procrastination." *Personality and Individual Differences* 35: 1401–1418.
- Wesp, R. 1986. "Reducing Procrastination through Required Course Involvement." *Teaching of Psychology* 13: 128–130.
- Wolters, C. A. 2003. "Understanding Procrastination from a Self-regulated Learning Perspective." *Journal of Educational Psychology* 95: 179–187.